

Cellular mechanisms of ocular disease

Part 1 | Disease mechanisms occupy different biological levels

Gene expression, cell survival and tissue architecture can fail separately.



A preserved upstream level does not prove every downstream level

Sequence, transcript, protein, cell and tissue are linked but distinct levels of evidence.

Level

Feature	Useful evidence	Interpretive limit
DNA	Sequence/copy number	Not expression alone
RNA/protein	Abundance or activity	Not complete tissue function
Cell/tissue	Morphology and architecture	Not one universal cause

Complement connects recognition to effectors

Different activation routes converge on a shared cleavage system.

Common downstream sequence

- C3 convertase cleaves C3
↓
- C3b supports opsonisation and convertase assembly
↓
- C5 cleavage yields inflammatory C5a and C5b
↓
- C5b initiates terminal membrane-attack assembly

Why target level matters

A proximal intervention at C3 can alter several downstream functions, whereas a terminal intervention leaves more upstream opsonising activity intact. Complement contributes both host protection and tissue injury; suppression of one function is not evidence that every branch of innate immunity is absent.

Adaptive responses have distinct outputs

Cellular activation, antibody secretion and local recruitment should not be collapsed into one label.

Process

Feature	Mechanistic level	Possible dissociation
T-cell activation	Signal-dependent transcription	Can change without B-cell depletion
Antibody production	B-cell/plasma-cell stages	Can persist after CD20 depletion
Graft sensitisation	Antigen drainage and node priming	Afferent route differs from effector access

Selectivity explains apparent contradictions

Detectable antibodies after B-cell depletion do not prove failure to remove CD20-positive cells. Reduced antigen-specific delayed hypersensitivity with preserved antibody can represent immune deviation. Measuring distinct outputs prevents a selective intervention being misclassified as universal immune paralysis.

Sources Truxima 500 mg Summary of Product Characteristics | The critical role of lymph nodes in corneal alloimmunization and graft rejection | Anterior Chamber-Associated Immune Deviation: An Acute Response to Ocular Insul...

Microbial mechanisms guide assay and drug interpretation

An organism's biology determines which target or growth condition is relevant.

Antibacterial target

Feature	Example class	Mechanism level
Cell-wall synthesis	Beta-lactams	Peptidoglycan assembly
Ribosomal function	Macrolides or aminoglycosides	Protein synthesis
DNA processing	Fluoroquinolones	Topoisomerase function

Two separate questions

Resistance can involve target alteration, drug inactivation or changed access. Organism identification does not automatically establish susceptibility. Similarly, PCR detection of a sequence and recoverable growth answer different questions: nucleic acid can persist after viable burden falls, and a negative test can reflect the specimen or assay.

Cell death and neoplasia need causal evidence

A morphological endpoint can be reached through different intracellular pathways.

Apoptotic pathway

Feature	Upstream event	Convergence
Intrinsic	Mitochondrial permeabilisation; cytochrome c	Caspase cascade
Extrinsic	Death-receptor signalling	Caspase cascade
Cross-talk	Pathways interact	Not perfectly isolated systems

Tissue interpretation

A tumour marker supports a cellular phenotype; architecture supports invasion and extent. A granuloma supports a reaction pattern; organism-directed evidence helps establish cause. Keep these levels separate, just as caspase activation alone does not identify which upstream apoptotic trigger initiated the response.